Single-cell RNAseq Workshop

From FASTQ to unfiltered Seurat/SingleCellExperiment object

Ian Beddows

October 9, 2026

Single-cell genomics workshop · Day 1

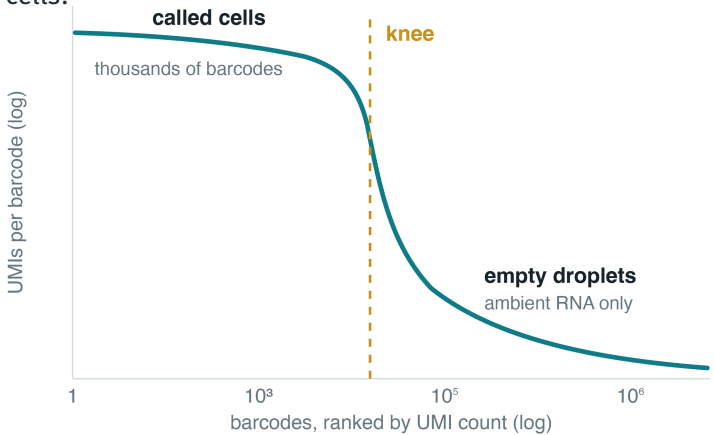
Quality Control

Quality Control

Post sequencing QC

- Sequence read quality
  - base quality v position
  - adapter content
  - duplication
  - detected barcodes/UMIs

What about neutrophils, resting lymphocytes, & red blood cells?



Mapping and counting features

Analysis-specific considerations

- Technology and **read architecture**

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- Reference genome (build/version)
- Gene annotation (source/version)

Gold-standard algorithms

RNA mapping (short-read single-cell)

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    - tool that turns salmon/piscem output to cell x gene matrix

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  - uses STAR

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  - custom barcode, UMI logic

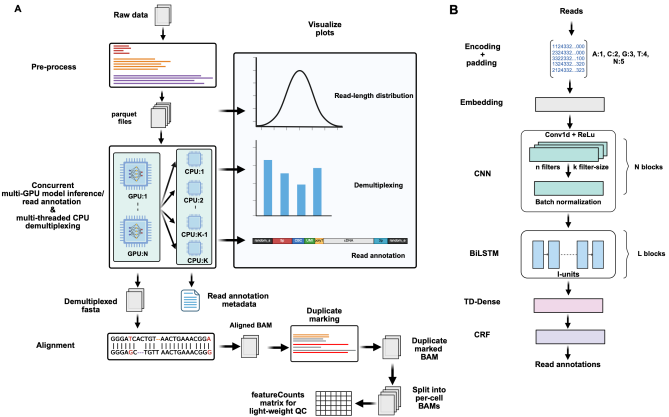
Gold-standard algorithms

RNA mapping (long-read single-cell)

- minimap2
  - Heng Li
- lr-kallisto

Read annotation

- Neural Network for long-read annotation and demultiplexing
  - Convolutional neural net to classify each nucleotide
  - Custom read architectures
  - Semwal et al. 2026.



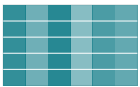
Post Alignment Library QC

- If you have a BAM, Picard RNAseq metrics
  - PCT_MRNA_BASES
  - PCT_RIBOSOMAL_BASES

Data Formats

Sparse Data

bulk: genes × 6 samples



single-cell: genes × ~20,000 cells (most entries zero)



Matrix Market Format

- MatrixMarket
  - Extension `.mtx`
  - Row, Column, Value
  - dgCMatrix R class used by Seurat and SingleCellExperiment
  - Must be read in full
- features.tsv.gz
- barcodes.tsv.gz

```
1    0      0       6      0
0   10.5    0       0      0
0    0    .015      0      0
0  250.5    0     -280    33.32
0    0      0       0      12

—
5  5  8
   1      1    1.000e+00
   2      2    1.050e+01
   3      3    1.500e-02
   1      4    6.000e+00
   4      2    2.505e+02
   4      4   -2.800e+02
   4      5    3.332e+01
   5      5    1.200e+01
```

Matrix Market From 10x Cell Ranger

Cellranger filtered features from zabrafish scRNAseq

```
[ian.beddows@compute002 filtered_feature_bc_matrix]$ ls
barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
[ian.beddows@compute002 filtered_feature_bc_matrix]$ zcat matrix.mtx.gz |head -6
%%MatrixMarket matrix coordinate integer general
%metadata_json: {"software_version": "cellranger-9.0.1", "format_version": 2}
32520 13176 21692115
378 1 1
813 1 1
[ian.beddows@compute002 filtered_feature_bc_matrix]$ zcat features.tsv.gz |head -2
ENSDARG00000103202  ENSDARG00000103202  Gene Expression
ENSDARG00000009657  fgfr1op2    Gene Expression
[ian.beddows@compute002 filtered_feature_bc_matrix]$ zcat barcodes.tsv.gz |head -2
AAACCAAAGCCTTCGC-1
AAACCAAAGCGCGATT-1
```

Hierarchical Data Format

- .h5, .hdf5
- Works with datasets larger than your RAM
- In R
  - HDF5Array and DelayedArray packages
  - Seurat and SingleCellExperiment read h5 directly
- Python: h5py and AnnData backed mode

Analysis in Python

Python

- scanpy
- Scales to larger datasets >1M cells
- faster
- AnnData objects
- bioconda

```
import scanpy as sc

# From .mtx
adata = sc.read_10x_mtx("outs/filtered_feature_bc_matrix/",
                        var_names="gene_symbols", cache=True)
# Or from .h5
adata = sc.read_10x_h5("outs/filtered_feature_bc_matrix.h5")
```

Analysis in R

Load Rstudio

Ondemand RStudio

Select R Version: **bbc2/R/alt/R-4.6.0-setR_LIBS_USER**

- R-4.6.0-setR_LIBS_USER has R packages pre-installed

Load Bioconductor Manager

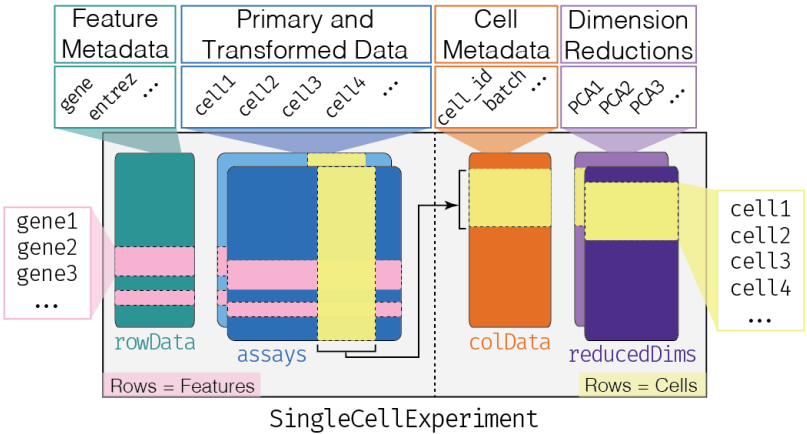
Bioconductor

```
if (!requireNamespace("BiocManager", quietly = TRUE)) {
  install.packages("BiocManager")
}
library(BiocManager)

# install packages with BiocManager
# BiocManager::install('<package_name>')
```

Single-cell experiment (SCE)

- Subclass that extends SummarizedExperiment S4 class
  - SummarizedExperiment functions work with SCE
  - reducedDims, altExps
  - SCE documentation



SCE packages

- scuttle: Computes per-cell QC metrics and does basic normalization and log-transformation.
- scater: Plots QC and expression, and runs PCA, t-SNE, and UMAP.
- scran: Provides pooled size-factor normalization, highly variable gene selection, and marker detection.
- bluster: Offers a unified interface for graph-based, k-means, and hierarchical clustering, plus diagnostics.
- batchelor: Corrects batch effects and integrates datasets, mainly with fastMNN.
- SingleR: Annotates cell types automatically against reference data.
- celldex: Supplies curated reference datasets for SingleR.
- scDblFinder: Detects doublets.
- BiocSingular: Runs fast approximate PCA/SVD on large datasets.
- BiocNeighbors: Performs fast nearest-neighbor searches for graph building and UMAP.
- slingshot: Infers trajectories and pseudotime.
- edgeR

SCE data import

```
# DropletUtils: Imports 10x data and calls empty droplets with emptyDrops.
library(DropletUtils)

# use filtered not raw features
sce <- DropletUtils::read10xCounts("outs/filtered_feature_bc_matrix/")
# read10xCounts also accepts the .h5 file
```

Seurat

Seurat vignettes

Install

```
if (!requireNamespace("Seurat", quietly = TRUE)) {
  BiocManager::install("Seurat")
}
library(Seurat)
```

Seurat object

- S4 container for single-cell data (`SeuratObject` package)
- Holds raw counts, normalized data, metadata, and analysis results together
- Created from a count matrix

Seurat object main slots

Slot	Contents
`assays`	Expression data (e.g. `RNA`)
`meta.data`	Per-cell annotations (nCount, nFeature, clusters)
`reductions`	PCA, UMAP, t-SNE embeddings
`graphs`	SNN / KNN graphs used for clustering
`active.ident`	Current cell identities (clusters)
`commands`	Log of Seurat functions run

Accessing Seurat data

```r
dim(seu)                  # genes x cells
head(seu@meta.data)       # cell metadata
seu$percent.mt <- PercentageFeatureSet(seu, pattern = "^MT-") # get MT pct

Idents(seu)               # active identities
Embeddings(seu, "umap")

colnames(seu)
Cells(seu)

Features(seu)
rownames(seu)

# List of object layers
Layers(seu)

# working with multimodal objects list assays
Assays(seu)

# Assay-specific features
Features(seu[["RNA"]])

# Variable feature names
VariableFeatures(seu)
```

Convert between Seurat & SingleCellExperiment

```
# Seurat -> SCE
sce <- Seurat::as.SingleCellExperiment(seu)

# SCE -> Seurat
seu <- SeuratObject::as.Seurat(sce, counts = "counts", data = "logcounts")
# as.Seurat imported with Seurat
```

AnnotationDbi

- AnnotationDbi
  - organism and TxDb
  - org.Hs.eg.db and TxDb.Hsapiens.UCSC.hg19.knownGene

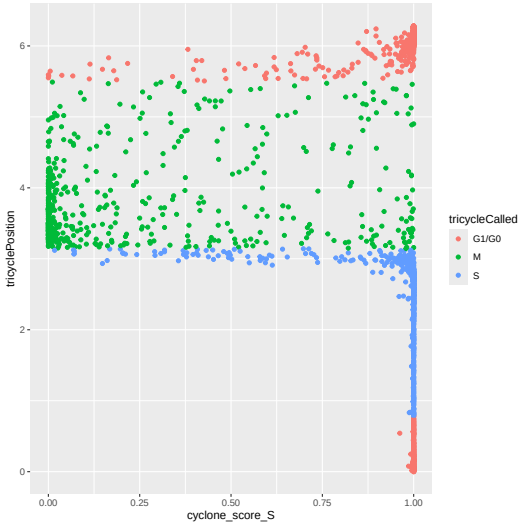
```
# map sce ENS genes to symbol
library(org.Hs.eg.db)
ens <- rownames(sce)
rowData(sce)$symbol  <- mapIds(org.Hs.eg.db,
                               keys      = ens,
                               column    = "SYMBOL",
                               keytype   = "ENSEMBL",
                               multiVals = "first")
```

Molecular Signatures

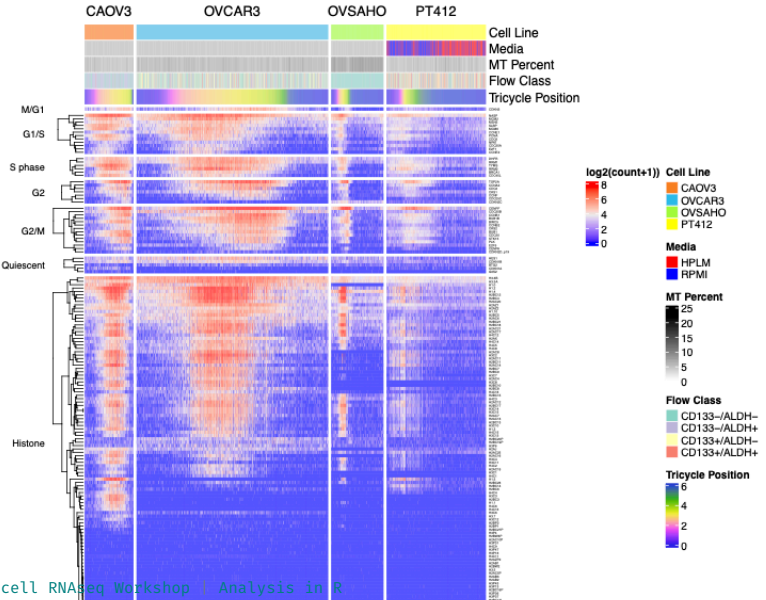
- Molecular Signatures DataBase
  - 9 major collections
- msigdbr R package
  - provides Molecular Signatures Database (MSigDB) gene sets
- ssGSEA to assign pathway score to all cells

Cell Cycle Estimation

1. Tricycle
   - estimates cell-cycle embedding using a fixed reference dataset
   - Article
   - Vignette
2. cyclone
   - scran package function
   - I do not recommend
3. Seurat regression
   - markers to estimate cell cycle then regress this out



Tricycle


CAOV3
OVCAR3
OVSAHO
PT412
Cell Line
Media
MT Percent
Flow Class
Tricycle Position
M/G1
G1/S
S phase
G2
G2/M
Quiescent
Histone
log2(count+1))
8
6
4
2
0
Cell Line
CAOV3
OVCAR3
OVSAHO
PT412
Media
HPLM
RPMI
MT Percent
25
20
15
10
5
0
Flow Class
CD133−/ALDH−
CD133−/ALDH+
CD133+/ALDH−
CD133+/ALDH+
Tricycle Position
6
4
2
0

Cell Annotation

- CELLxGENE
- Allen Brain Cell Atlas
- Azimuth
  - `Seurat::RunAzimuth()`
- scGate
  - uses known marker genes

Integration

- scRNA (batch correction)
  - Harmony
    - Returns corrected PC embeddings
    - Iterative soft k-means clustering
  - batchelor
    - Returns corrected PC embeddings & counts
    - fastMNN
- Multi-omics
  - MOFA2
    - Factor analysis general framework for the integration of multi-omic data sets

Rest of today

Plan

- Load MatrixMarket data into a Seurat Object
- Include metadata in Seurat mata.data slot
- Explore object

Load MatrixMarket as dgCMatrix

```
library(Seurat)
library(SingleCellExperiment)

base    <- "/Volumes/projects/bbc/research/workshop_Oct2026_Kin/data"
samples <- paste0("sample", 5:10)

# One sparse matrix
counts <- do.call(cbind, lapply(samples, function(s) {
  m <- Seurat::Read10X(file.path(base, s))
  colnames(m) <- paste(s, colnames(m), sep = "_")
  m
}))

dim(counts)
```

```
[1] 29453 20935
```

dgCMatrix to Seurat

```r
cell_meta <- read.csv(file.path(base, "per_cell_meta.csv"), row.names = 1)

rownames(cell_meta) <- paste0("sample", cell_meta$sample, "_", cell_meta$barcode)

cell_meta <- cell_meta[colnames(counts), ]

seu <- CreateSeuratObject(counts, meta.data = cell_meta, project = "workshop")
seu
```

```
An object of class Seurat
29453 features across 20935 samples within 1 assay
Active assay: RNA (29453 features, 0 variable features)
 1 layer present: counts
```

dgCMatrix to SingleCellExperiment

```
sce <- SingleCellExperiment(list(counts = counts), colData = cell_meta)
sce
```

```
class: SingleCellExperiment
dim: 29453 20935
metadata(0):
assays(1): counts
rownames(29453): Xkr4 Gm1992 ... CAAA01147332.1 tomato-td
rowData names(0):
colnames(20935): sample5_AAACCTGAGACTGTAA sample5_AAACCTGAGATGCCTT ...
  sample10_TTTGTCATCATTATCC sample10_TTTGTCATCCCATTTA
colData names(10): barcode sample ... doub.density sizeFactor
reducedDimNames(0):
mainExpName: NULL
altExpNames(0):
```

Seurat object structure

```
str(seu, max.level = 2)
```

```
Formal class 'Seurat' [package "SeuratObject"] with 13 slots
  ..@ assays      :List of 1
  ..@ meta.data   :'data.frame':    20935 obs. of  13 variables:
  ..@ active.assay: chr "RNA"
  ..@ active.ident: Factor w/ 6 levels "sample10","sample5",..: 2 2 2 2 2 2 2 2 2 ...
  .. ..- attr(*, "names")= chr [1:20935] "sample5_AAACCTGAGACTGTAA" "sample5_AAACCTGAGA
  ..@ graphs      : list()
  ..@ neighbors   : list()
  ..@ reductions  : list()
  ..@ images      : list()
  ..@ project.name: chr "workshop"
  ..@ misc        : list()
  ..@ version     :Classes 'package_version', 'numeric_version'  hidden list of 1
  ..@ commands    : list()
```

Seurat object structure

```
Seurat::GetAssayData(seu, layer = "counts")[1:5, 1:5]
```

```
5 x 5 sparse Matrix of class "dgCMatrix"
       sample5_AAACCTGAGACTGTAA sample5_AAACCTGAGATGCCTT
Xkr4                          .                        .
Gm1992                        .                        .
Gm37381                       .                        .
Rp1                           .                        .
Sox17                         .                        .
       sample5_AAACCTGAGCAGCCTC sample5_AAACCTGCATACTCTT
Xkr4                          .                        .
Gm1992                        .                        .
Gm37381                       .                        .
Rp1                           .                        .
Sox17                         .                        .
       sample5_AAACGGGTCAACACCA
Xkr4
```